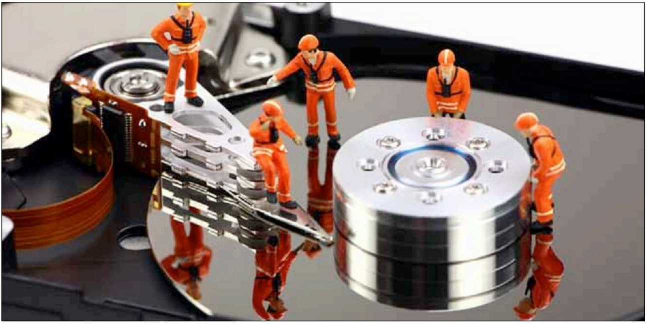
A Look at the Basics of LVM

This article deals with Logical Volume Management (LVM) and looks at how to set it up.



Many years ago, when 2 GB of storage was considered a lot, I beamed with pride after installing two hard disks, each 2 GB. With 4 GB, I was sure of never running low on space. Such pride is shortlived, as we know. Down the road, it was a limitation when I wanted to install more software into my Linux distro (those were the days when Red Hat Linux was the thing to have) and the disk that had my root partition just didn't have the space. There had to be some way to resolve the issue; after all, my system had a total of four gigs, right? Was there a way I could, well, 'combine' the two drives? Yes, there was: three letters of pure genius, LVM. Logical Volume Management, put simply, is a concept that combines physical drives and makes them appear as contiguous spaces.

The basics of LVM

LVM is a virtual layer that sits sandwiched between the operating system and your storage devices, i.e., hard disks. It creates groups of your hard disks, called Volume Groups (VGs). Within these VGs, you can create partitions, called Logical Volumes (LVs). The operating system will never know the difference; the behaviour is the same as accessing any regular disk/partition. The benefits of LVM over regular filesystems include: dynamic resizing of VGs, and live snapshots of partitions without unmounting them. Let's now jump right in to how to set up LVM on our system. The OS I'm using for this is Ubuntu 12.10. There are two scenarios in which one would set up LVM: while performing a Linux install, or post-installation. LVM can be configured using command-line and GUI tools. I shall walk you through all of this.

Figure 1: What LVM actually does (is)

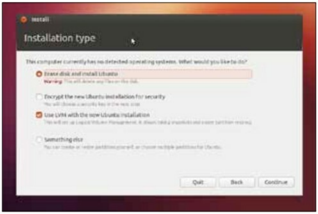


Figure 2: Starting LVM partitioning